

AQA (Trilogy) Combined Science

GCSE Unit 6.2 Electricity

Test (Levels 4 – 9)

Time allowed: **50 minutes**

Question	Links to Student Progress Sheet	Score
1	6.2.1.1.a, 6.2.1.2.a	
2	6.2.4.1.b, 6.2.4.2.e	
3	6.2.1.3.d, 6.2.1.4.a	
4	6.2.1.1.a, 6.2.1.4	
5	6.2.4.3	
6	6.2.2, 6.2.3.2	
7	6.2.1.4.e	
8	6.2.1.4, 6.2.1.3	
Total	XX	

Score	Estimated Grade
47 – 50	9
40 – 46	8
33 – 39	7
28 – 32	6
24 – 27	5
20 – 23	4
0 – 19	U

Please be aware that these grade boundaries are Twinkl's own, and may not be in-line with AQA's

own.Area for Improvement:

1.

a. Match up the diagram of the circuit component
with the correct label:



lamp

fuse



LDR thermistor

[3 marks]



diode

[1 mark] 4

b. Electrical charge is transferred by what in
metals?

2. The table below shows the power of different electrical appliances.

Appliance					
Power (w)	500	1500	1600	2500	700

A student wants to use all of the appliances at the same time so plugs them in to a multiway socket.

The socket has a maximum safe current of 30A and the mains electricity is 230V.

a. Is it safe to use all five appliances at once?

Explain your answer showing your use of the equation:

Current = power $\div$ potential difference

[4 marks]

- b. The student wants to boil the kettle. It takes 150 seconds for 2400 coulombs of charge to pass through it.

Calculate the current flow through the kettle's heating element and give the correct unit.

Current = unit =

[3 marks] 7

Page 3 of 13

3.

- a. What is the equation which links resistance, current and potential difference?

Electricity **Test**

[1 mark]

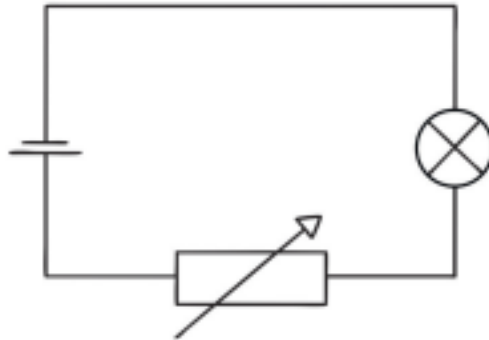
- b. Using the equation from part a. calculate the resistance when a lamp has a current

of 0.17A and a potential difference of 3.6V?

Resistance = Ω

[3 marks]

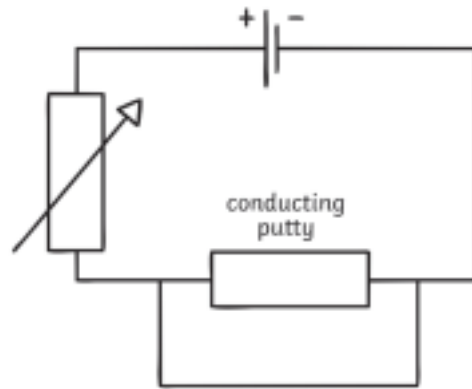
c. A circuit is set up using a variable resistor as shown below.



What happens to the bulb as the variable resistor is increased and decreased? **[2 marks] 6**

4. A student set up the circuit below to investigate the resistance of a piece of conducting putty, but forgot to put in the voltmeter and the ammeter.

a. Draw the correct symbols for a voltmeter and an ammeter in the correct places on the circuit.

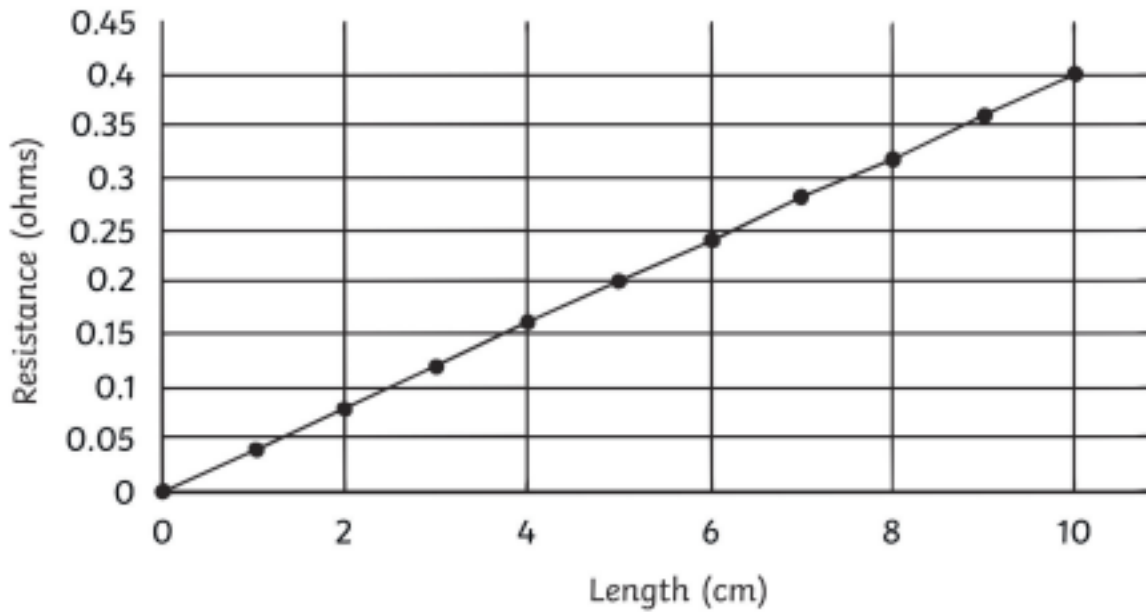


[2 marks]

b. In a circuit the current can be changed. Explain how.

[2 marks]

The conducting putty is rolled into cylinders which have the same thickness but are different lengths. The graph below shows how length affects resistance.



c. Using the graph above, find the resistance at 6cm

Resistance = ohms

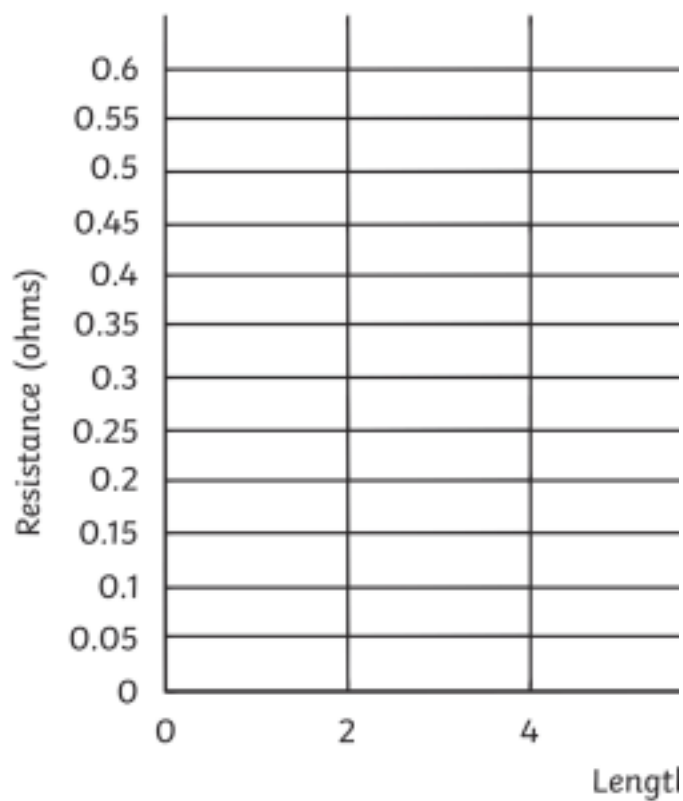
[1 mark]

d. The current at 6cm was 0.15A. Use your answer to part c. to calculate the potential difference at 6cm. Show your working out.

Potential difference = volts

[2 marks]

A thicker piece of putty was tested and the graph below shows the resistance.



Thickness (cm)e. What is the relationship between the thickness of the putty and the resistance?
Electricity **Test**

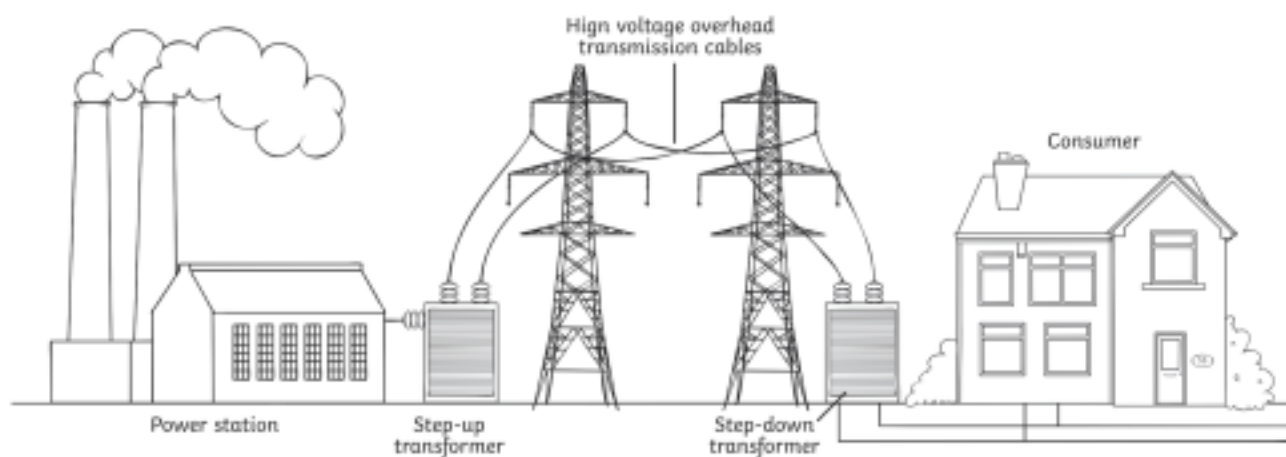
[1 mark]

f. Give an example of one error that could have been made which may cause the results to be inaccurate?

9

[1 mark]
H

5. The National Grid transfers electrical power to consumers from power stations.
The diagram below shows parts of that system.



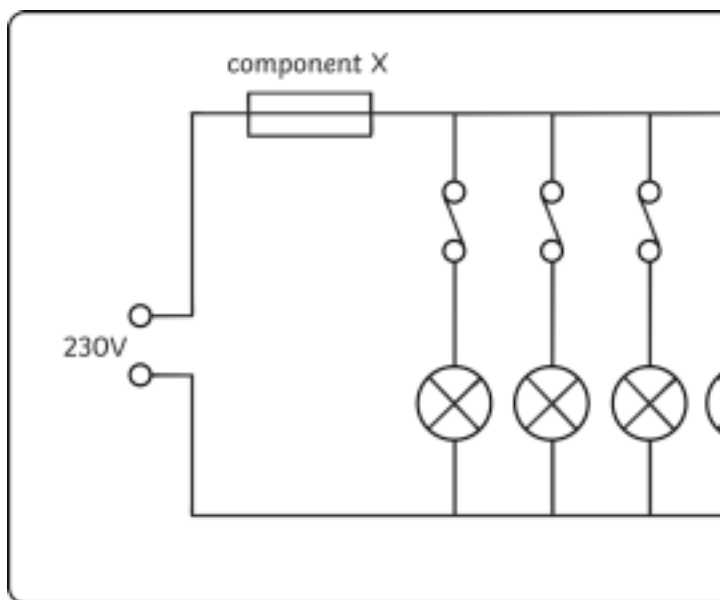
- a. Why are step-up transformers used in the National Grid?

[3 marks]

- b. What does the step-down transformer do?

[1 mark] 4

6. The diagram shows how the lights in a house



a. How is this circuit different from a series circuit?

b. Why is it better to have a parallel circuit than a series circuit for the lighting in a house?

[1 mark]

c. What is component X?

Electricity **Test**

[1 mark]

[1 mark]

- d. Lamp Y was replaced by a low energy lamp which uses less power but provides the same amount of light. Which row of the table shows the right comparison of lamp Y to the low energy lamp?

Tick **one** box

	Current in low energy lamp when compared to the current in the lamp Y	Voltage across low energy lamp when compared to the voltage across lamp Y
A	Same as	Less than
B	Less than	Same as
C	Same as	Same as
D	Less than	Less than

[1 mark]

- e. The mains supply is alternating current. Explain what alternating current is.

[2 marks]

- f. What is the frequency of mains alternating current?

mark]
7

[1

7.

a. Thermistors are used in many situations. Which of the following could be one?

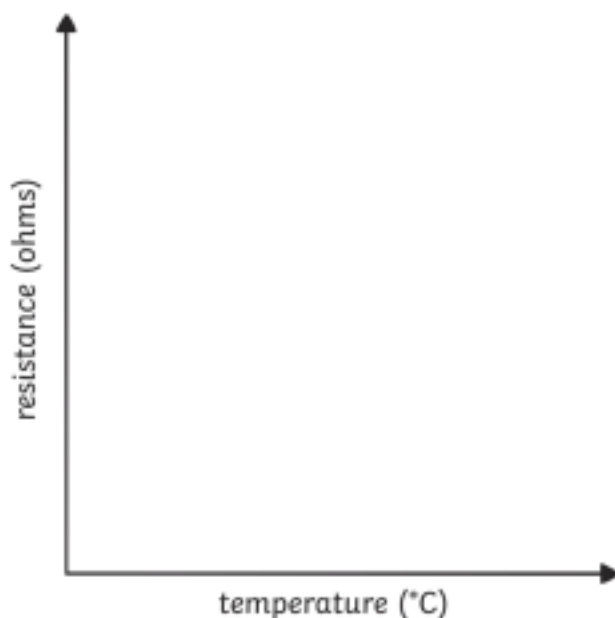
When it's dark, an automatic circuit which turns a light on outside

A circuit which turns on automatically to water plants

A circuit which automatically turns a heating system on and off

[1 mark]

b. As the temperature changes, the resistance of a thermistor changes with it. Sketch the shape of the graph to show what it would like.

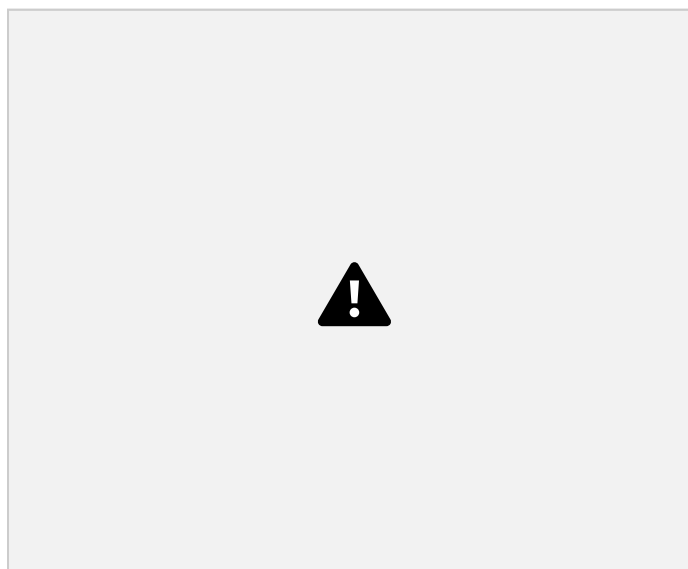


[1 mark]

c. Draw the circuit symbol for a thermistor.

[1 mark]

8. A student carried out an experiment using a filament bulb and set up the following circuit to measure the current and potential difference.



- a. What is component C and why is it in the circuit?

[1 mark]

- b. As the potential difference increases, so does the resistance of the bulb, why?

[1 mark]

- c. At 9V, the bulb is at full brightness. At this point, the power through the bulb is 2.5A
Calculate the power of the bulb when it is at full brightness – give the unit.

Power = unit =

[2 marks]

d. The following table shows two different types of bulb and data relating to each.

Bulb type	Energy efficiency	Cost of bulb	Average lifetime in hours
LED (Light Emitting Diode)	34%	£12.20	36 000
Halogen	12%	£2.05	2 000

Both bulbs give the same amount of light. Compare the two bulb types relating to their energy efficiency and cost.

[6 marks] 10

End of Test